



# MIET AFRICA

Youth Skills Centre Platform

**Group 5**

Work Integrated Learning Project Part 1

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## Table of Contents

|                                                                             |          |
|-----------------------------------------------------------------------------|----------|
| <b>Research and planning .....</b>                                          | <b>2</b> |
| <b>Explanation of the research done into the sourcing of a client .....</b> | <b>2</b> |
| <b>Research Conducted .....</b>                                             | <b>2</b> |
| <b>Description of the client.....</b>                                       | <b>3</b> |
| <b>Client requirements .....</b>                                            | <b>3</b> |
| • <b>Functional Requirements .....</b>                                      | <b>3</b> |
| • <b>Non-Functional Requirements .....</b>                                  | <b>5</b> |
| <b>Background information on the system.....</b>                            | <b>6</b> |
| <b>System high level design.....</b>                                        | <b>6</b> |
| <b>System Architecture .....</b>                                            | <b>7</b> |
| <b>System Design .....</b>                                                  | <b>7</b> |
| <b>UML diagrams .....</b>                                                   | <b>8</b> |
| <b>Entity Relationship diagrams .....</b>                                   | <b>8</b> |

## Research and planning

We began by reviewing the Borgen Project's published directory of South African NGOs. After evaluating ten organisations, we selected MIET Africa because of their direct focus on youth skills development in under-resourced communities a problem domain where technology can deliver measurable, scalable impact. Specifically, their Youth Skills Centre programme presented a clear need for a learner management system that both a desktop application (for administrators) and a mobile application (for facilitators) could directly address.

## Explanation of the research done into the sourcing of a client

To find a suitable client for this project, research was carried out on several non-governmental organisations operating in South Africa, particularly those involved in education and youth development. After reviewing different organisations, MIET Africa was selected because of its strong focus on supporting children and young people in under-resourced communities. Information was gathered from the organisation's official website and other publicly available sources to better understand how they operate and the challenges they face. The research showed that many of their processes are still manual and paper-based, which makes it difficult to manage learner information, track progress, and report results to donors. These findings helped justify the need for a digital system and guided the design of the proposed solution for this project.

## Research Conducted

- Reviewed MIET Africa's official website ([www.mietafrica.com](http://www.mietafrica.com)) and programme documentation.
- Studied the Borgen Project's NGO profile listing for MIET Africa.
- Reviewed SETA (Sector Education and Training Authority) compliance requirements for learner records in South Africa.
- Researched POPIA (Protection of Personal Information Act 4 of 2013) obligations for storing personal data, including that of minors.

- Reviewed similar learner management systems deployed by South African Skills Centres supported by the National Skills Fund (NSF).
- Consulted academic literature on digital transformation in the non-profit sector and NGO information systems in developing countries.

## Description of the client

| Field                | Details                                                                             |
|----------------------|-------------------------------------------------------------------------------------|
| Organisation Name    | MIET Africa                                                                         |
| Type                 | Non-Governmental Organisation (NPO/NGO)                                             |
| Founded              | 1990                                                                                |
| Headquarters         | Durban, KwaZulu-Natal, South Africa                                                 |
| Website              | www.mietafrica.com                                                                  |
| Primary Focus        | Education & Youth Skills Development                                                |
| Programme in Scope   | Youth Skills Centre Programme                                                       |
| Target Beneficiaries | Youth aged 15–35 in under-resourced communities across South Africa                 |
| Geographic Reach     | KwaZulu-Natal, Limpopo, Eastern Cape (primary), wider sub-Saharan Africa            |
| Languages Served     | English, isiZulu, isiXhosa, Afrikaans, Sesotho                                      |
| Legal Compliance     | POPIA (Act 4 of 2013), SETA regulations, NPO Act                                    |
| Key Partners         | Government departments, schools, UNICEF, National Skills Fund, international donors |

## Client requirements

### • Functional Requirements

Functional requirements define what the system must do the specific features and behaviours it must provide to its users such as:

### ***Learner Management***

| Functional Requirement                                                                                                                           | User Role       |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Admin can register a new learner with: full name, ID number, date of birth, gender, contact number, physical address, and programme of interest. | Admin           |
| System must display a searchable and filterable learner directory (by name, ID, course, province, status).                                       | Admin / Manager |
| Admin can edit an existing learner profile or deactivate a learner without deleting their records.                                               | Admin           |
| System must link each learner to one or more enrolled courses automatically.                                                                     | Admin           |
| Learner profile must show full enrolment history, attendance percentage, assessment results, and certificate status.                             | Admin / Manager |

## Course and Attendance Management

| Functional Requirement                                                                                                          | User Role       |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Admin can create and manage courses with: title, description, duration (hours), facilitator assigned, start date, and end date. | Admin           |
| Facilitator can mark attendance per learner per session on the iOS mobile app, including when offline.                          | Facilitator     |
| System must automatically calculate and display attendance percentage per learner per course.                                   | System (auto)   |
| System must flag and alert admin when a learner's attendance drops below the 80% threshold.                                     | System (auto)   |
| Admin can generate attendance reports filtered by course, date range, facilitator, or individual learner.                       | Admin / Manager |

## Assessment and Certification

| Functional Requirement                                                                                                                  | User Role       |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| System must automatically calculate a final result (Pass / Fail) based on a configurable pass mark set per course.                      | System (auto)   |
| Admin can generate a branded PDF certificate for learners who have successfully passed a course.                                        | Admin           |
| Generated certificate must include: learner full name, course title, completion date, MIET Africa logo, and authorised signature block. | System (auto)   |
| System must maintain a complete assessment history per learner, viewable over time.                                                     | Admin / Manager |

## Reporting and Impact Tracking

| Functional Requirement                                                                                                        | User Role       |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------|
| System must auto-generate monthly and quarterly impact reports based on all captured data.                                    | System (auto)   |
| Reports must include: total learners registered, courses completed, pass rates, certificates issued, and attendance averages. | Admin / Manager |
| Admin can export all reports to both PDF and CSV format for external submission.                                              | Admin           |
| A donor-facing portal must display key impact statistics in real time without login required.                                 | Donor (public)  |
| System must send automated email notifications to learners when they pass a course or receive a certificate.                  | System (auto)   |

## User and System Administration

| Functional Requirement                                                                                    | User Role     |
|-----------------------------------------------------------------------------------------------------------|---------------|
| IT Admin can create, edit, and deactivate system user accounts for any role.                              | IT Admin      |
| System must enforce role-based access control (RBAC) for: Admin, Facilitator, Manager, and Donor roles.   | System        |
| All user actions (create, edit, delete, export) must be logged in a timestamped audit trail.              | System (auto) |
| Mobile app must support full offline data entry with automatic background sync when internet is restored. | Facilitator   |
| IT Admin can trigger a full system data backup and restore from backup when needed.                       | IT Admin      |

- **Non-Functional Requirements**

Non-functional requirements define how the system must perform its quality attributes, constraints, and compliance standards such as:

- **Security:** All passwords must be hashed using bcrypt. Authentication must use JWT tokens with a 24-hour expiry. It protects sensitive learner personal data from unauthorised access.
- **Data Privacy System** must comply with South Africa's POPIA (Act 4 of 2013) for all personal data stored, including learner ID numbers and contact details. Legal requirement when storing personally identifiable information of learners.
- **Performance Desktop app** must load any screen within 2 seconds. Mobile app must load its home screen within 3 seconds on a mid-range iOS device. Users in the field need responsive tools slow apps are abandoned.
- **Offline Support iOS** mobile app must support full offline functionality for attendance and assessment capture, with auto-sync when internet is restored. Many Skills Centre venues have unreliable or no connectivity.
- **Scalability:** System must support up to 5,000 learner records and 500 concurrent users without performance degradation. MIET Africa operates across multiple provinces, and the system must grow with them.
- **Usability:** The mobile app must be usable by a facilitator with no prior smartphone app experience within 15 minutes of first use. Facilitators are not technical users and need simple, intuitive interfaces.
- **Availability:** The cloud API and database must maintain 99.5% uptime with automated failover in place. Admin staff rely on the desktop app for daily, time-sensitive operations.
- **Compatibility:** Mobile app must support iOS 15 and above (SwiftUI). Desktop app must run on Windows 10/11 and macOS 12 and above. It ensures the system runs on devices available to MIET Africa staff.
- **Maintainability:** Codebase must follow documented coding standards with a minimum of 70% unit test coverage across all modules. It allows future student teams or MIET Africa developers to maintain and extend the system.
- **Accessibility:** UI must comply with WCAG 2.1 AA contrast standards. All font sizes must be user-adjustable in settings. It ensures no learner or facilitator is excluded due to visual limitations.

## Background information on the system

South Africa faces a severe youth unemployment crisis, with official unemployment among those aged 15–34 exceeding 45% in 2025. NGOs like MIET Africa play a critical role in equipping young people with the vocational and life skills needed to access employment. However, without a digital management system, MIET Africa's Youth Skills Centre cannot scale its operations, prove impact to donors, or maintain learner records that comply with SETA and POPIA regulations. The current manual process is not sustainable. As more learners enrol and more courses are offered, the volume of paper records becomes unmanageable. Staff spend hours on administrative tasks that a well-designed system could automate in seconds time that could instead be invested in supporting learners directly.

## System high level design

The system consists of three interconnected components: a Desktop Application for administrative users, an iOS Mobile Application for facilitators working in the field, and a Cloud Backend (REST API + database) that both applications communicate with. The mobile app also supports a full offline mode using local device storage (Core Data), syncing automatically when connectivity is restored.

## System Architecture

| System Layer         | Technology                             | Justification                                                                                                            |
|----------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Desktop App Frontend | Electron.js + React                    | Cross-platform desktop app (Windows & macOS) built from a single web-technology codebase                                 |
| Mobile App Frontend  | SwiftUI (iOS 15+)                      | Native iOS app providing optimal performance, full Apple ecosystem integration, and excellent accessibility support      |
| Backend API          | Node.js + Express.js                   | RESTful API handling all data requests from both apps; lightweight, scalable, and well-documented                        |
| Cloud Database       | PostgreSQL (AWS RDS)                   | Robust relational database ideal for the structured learner, course, attendance, and assessment data model               |
| Local Mobile Storage | Core Data (iOS)                        | Provides full offline capability on the iOS app; syncs all queued records to the cloud API when connectivity is restored |
| Authentication       | JWT (JSON Web Tokens) + bcrypt         | Stateless, secure token-based authentication; passwords hashed with bcrypt before storage                                |
| File Storage         | AWS S3                                 | Secure cloud storage for generated PDF certificates and any uploaded learner supporting documents                        |
| Push Notifications   | Apple Push Notification Service (APNs) | Delivers push notifications to iOS devices for low-attendance alerts and course milestones                               |
| Email Notifications  | SendGrid API                           | Sends automated emails to learners on course completion and certificate issuance                                         |
| Cloud Hosting        | AWS EC2 + Load Balancer                | Scalable, reliable hosting with automated failover ensuring 99.5% uptime SLA                                             |
| Version Control      | GitHub                                 | Source code repository, issue tracking, documentation, and CI/CD pipeline integration                                    |

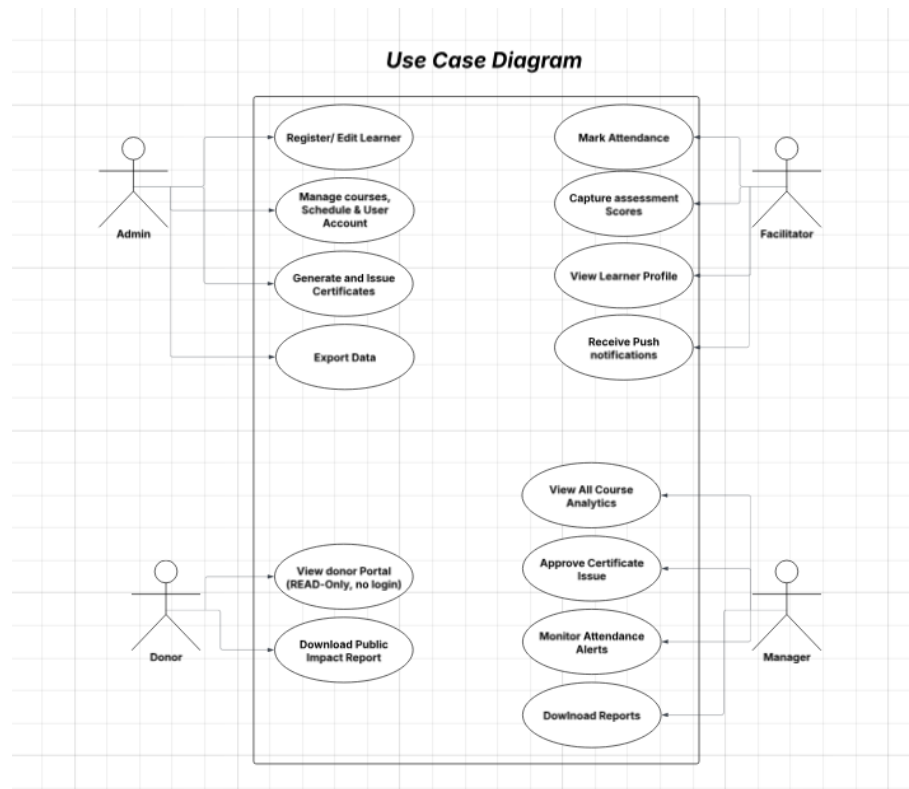
## System Design

The system was designed to provide an efficient and easy-to-use digital platform that supports the needs of MIET Africa and its stakeholders. The solution consists of multiple components working together, including a user interface for administrators and facilitators, a backend system that processes requests and manages the application logic, and a secure database that stores all learner and programme data. The design allows administrators to register learners, manage courses, track attendance, record assessments, and generate reports or certificates. The system also supports mobile access for facilitators so that information can be updated directly from the classroom. This overall design ensures that data is organised, accessible, and secure while improving the efficiency of MIET Africa's operations.



## UML diagrams

### Use Case Diagram



## Entity Relationship diagrams

